

02/12/20

Sorgenti di $\vec{B}$:
CORRENTI

simmetrie
 $\longrightarrow$
LEGGE
DI AMPÈRE

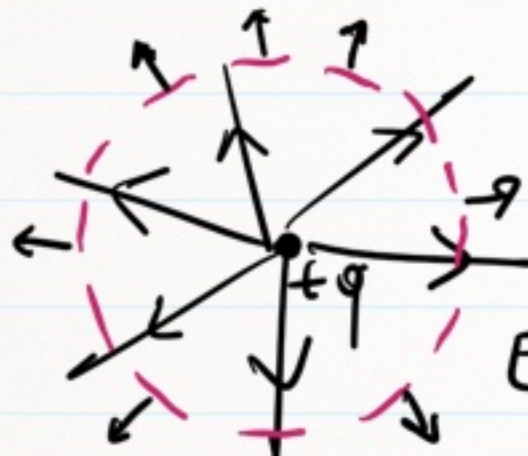
Determinare $\vec{B}$
nello spazio
 $(x, y, z) \equiv \vec{r}$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{conc}}$$

Σ chiusa

per $\vec{E}$, LEGGE DI
GAUSS



$$\oint \vec{E} \cdot d\vec{S} = q_{\text{int}}$$

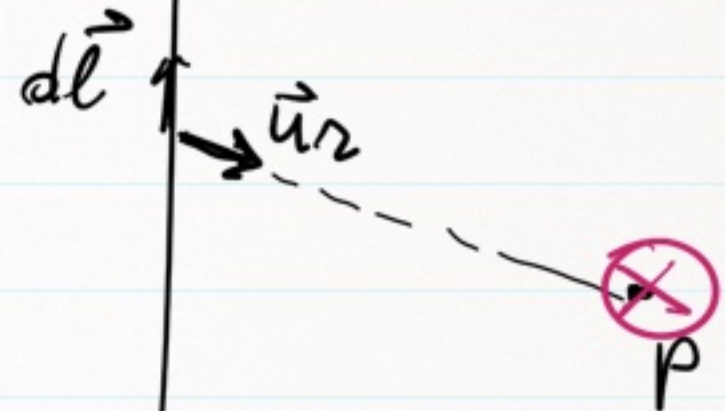
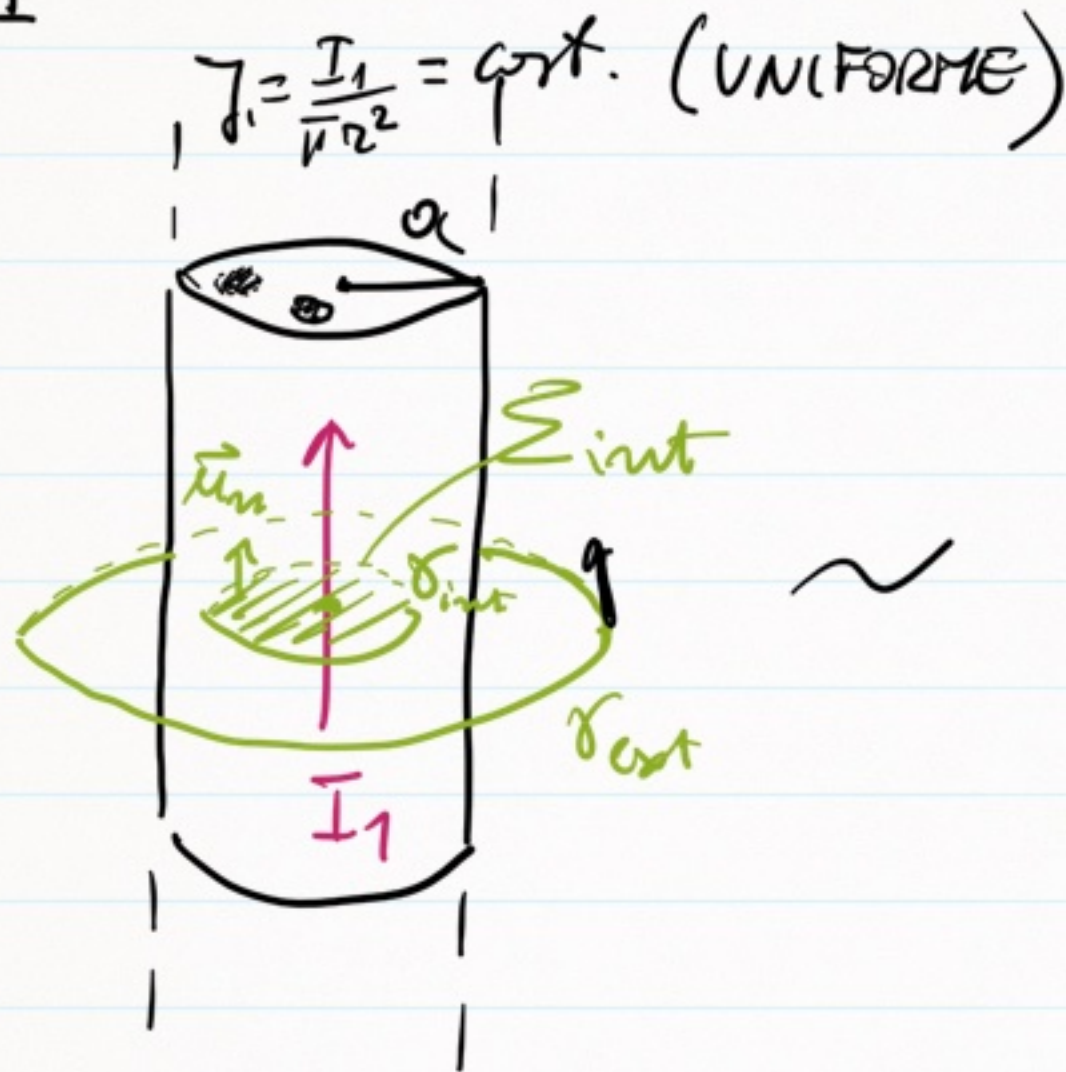
$$E \propto \frac{1}{r^2}$$

per $\vec{B}$ con AMPÈRE

cerchiamo circuiti
(di integrazione) su
cui $\vec{B}$ è costante
(TANGENTE o NORMALE)

Scheda 1

Ex. 5



$$d\vec{B}_P \parallel \vec{u}_\varphi$$

$$d\vec{\ell} \times \vec{u}_r = \vec{u}_\varphi$$

$$\underline{d\vec{B}_P} = \frac{\mu_0}{4\pi} I_1 \frac{d\vec{\ell} \times \vec{u}_r}{r^2}$$

$\int \dots$ Laplace

$$\oint_{\gamma} \vec{B} \cdot d\vec{S} = \mu_0 \iint_{\Sigma} \vec{J}_1 \cdot \vec{u}_n d\Sigma$$



$$0 \leq r < \infty$$

$$I_1$$

$$\left[\begin{array}{ll} - \vec{J}_1 \parallel \vec{u}_n & - \vec{B} \parallel d\vec{S} \\ - \vec{J}_1 \text{ uniforme su } \Sigma & - B = B(r) \end{array} \right]$$

$$\mu_0 J_1 \iint_{\Sigma} d\Sigma = B \oint_{\gamma} ds$$

$$\mu_0 J_1 \pi r^2 = B \cdot 2\pi r$$

$$B = \frac{\mu_0 J_1 r}{2} = \frac{\mu_0 I_1}{2\pi a^2} r$$

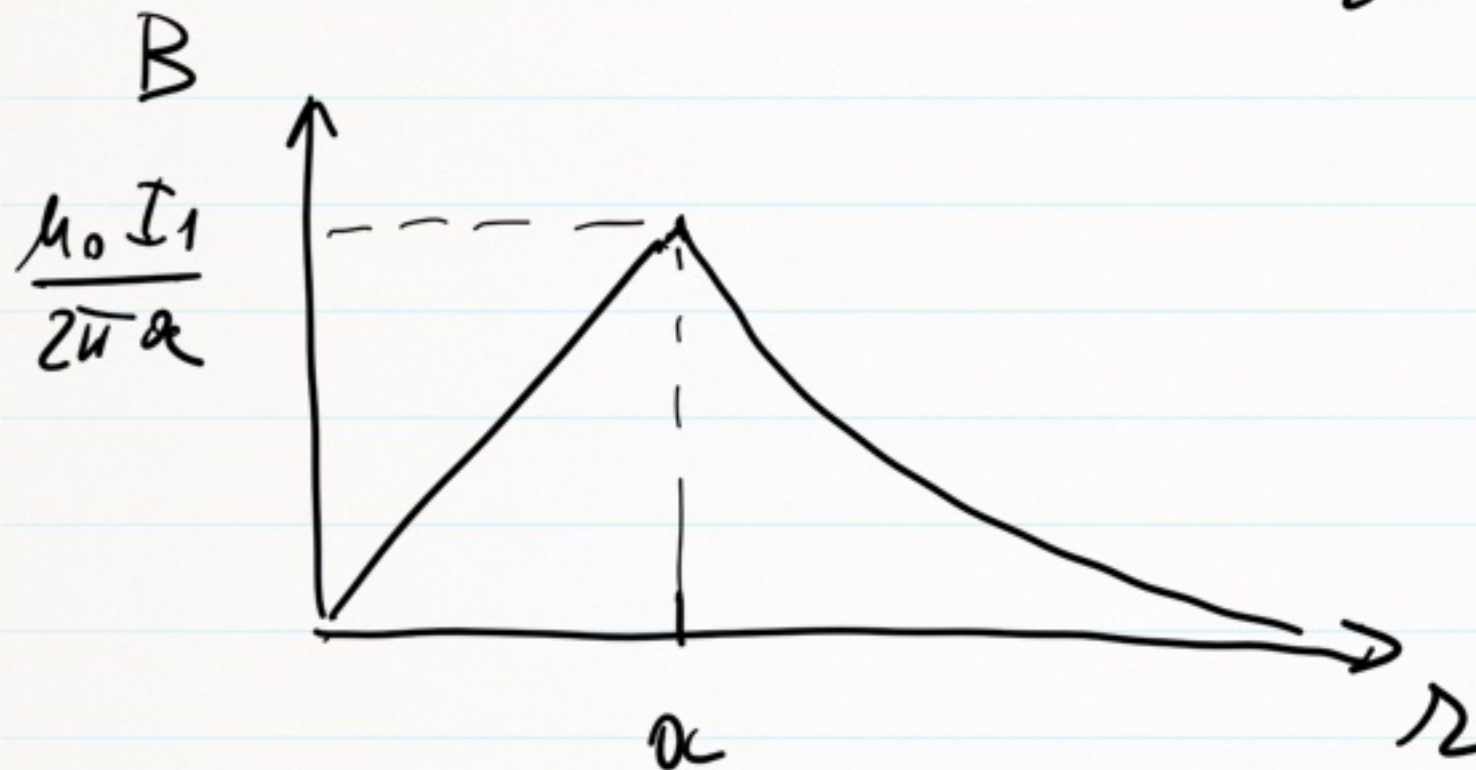
$$\bullet \quad 0 \leq r \leq a$$

$$B = \frac{\mu_0 I_1}{2\pi a^2} r$$

$$\bullet \quad a \leq r < \infty$$

$$B \cdot 2\pi r = \mu_0 I_1$$

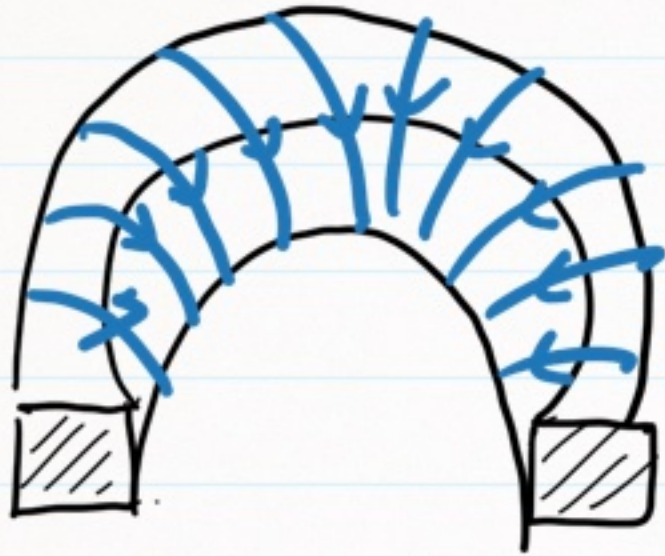
$$B = \frac{\mu_0 I_1}{2\pi r}$$



$$d\vec{F}_{12} = I_2 d\vec{\ell}_2 \times \vec{B}_1(z)$$

$$\vec{F}_{12} = \frac{\mu_0 I_2 I_1}{2\pi b} \underbrace{\oint_2 d\ell_2}_h = \frac{\mu_0 I_2 I_1 h}{2\pi b}$$

Ex. 8

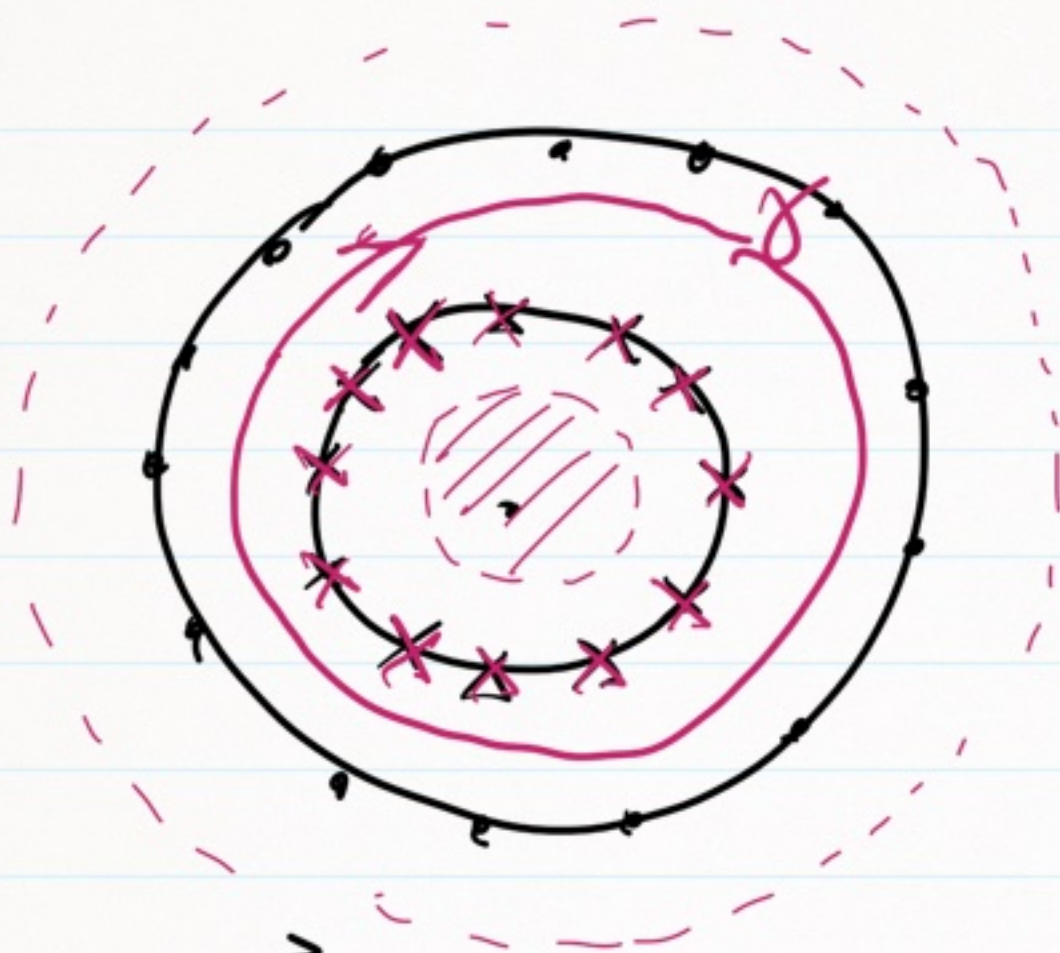


SOLENOIDE
TOROIDALE
(sezione quadrata)

$$B \approx \mu_0 n I \quad \text{solenoide } l \gg R$$

per un solenoide toroidale : $B \approx \mu_0 n I$

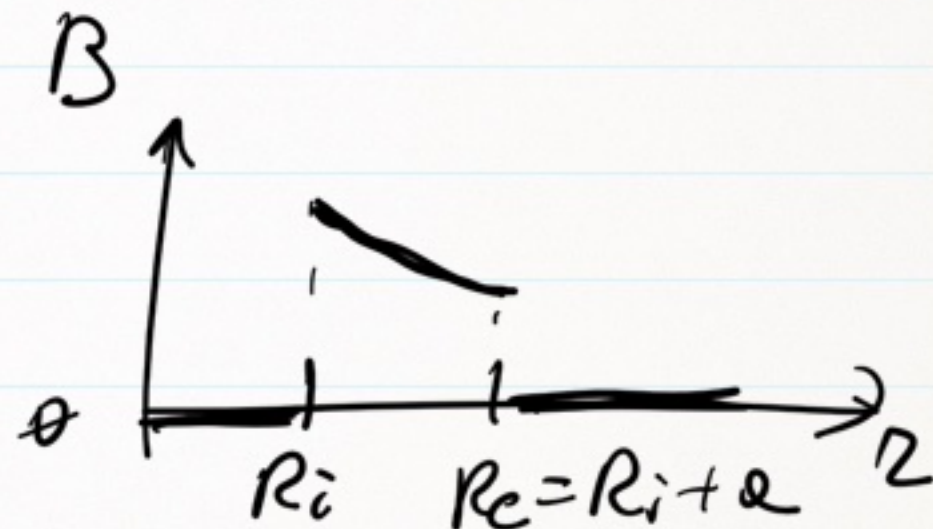
$$n \approx \frac{N}{2\pi R_m}, \quad R_m = \frac{R_i + R_e}{2}, \quad \text{se } R_m \gg a, \quad R_e = R_i + a$$



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \underbrace{I_{enc}}_{N \cdot I}$$

$$B \cdot 2\pi r = \mu_0 N I$$

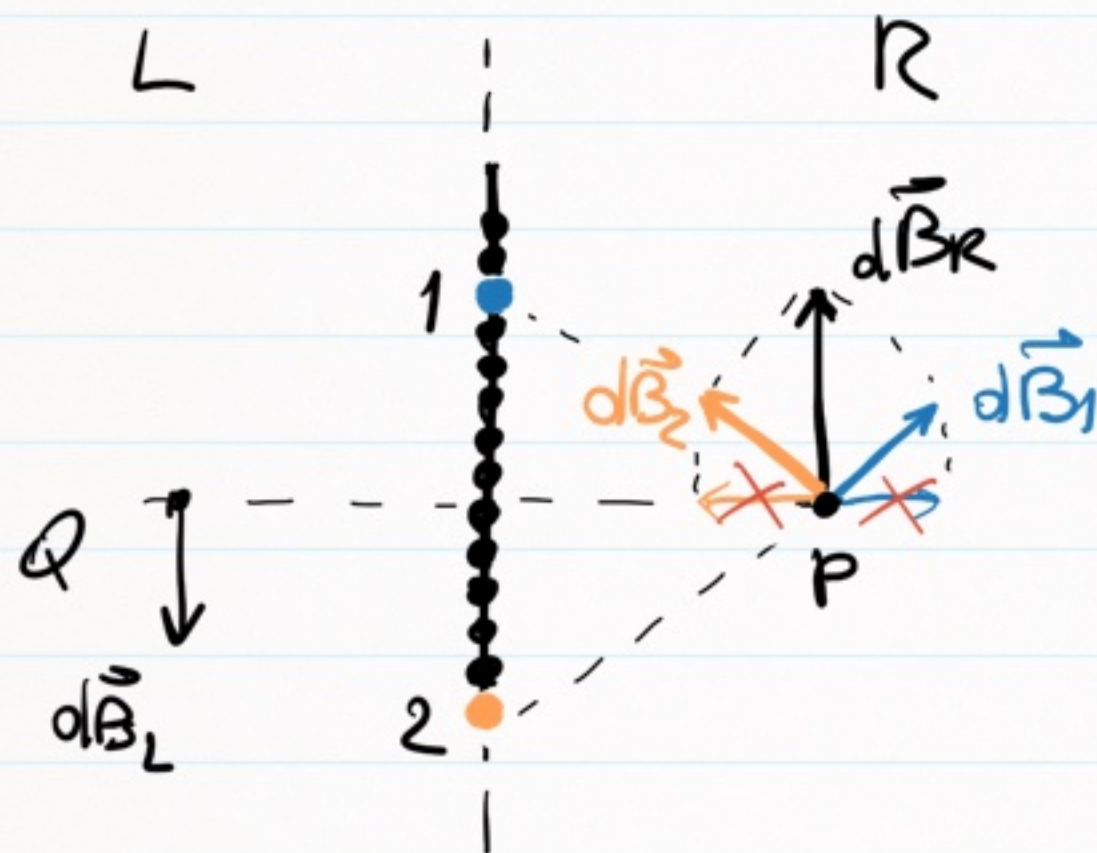
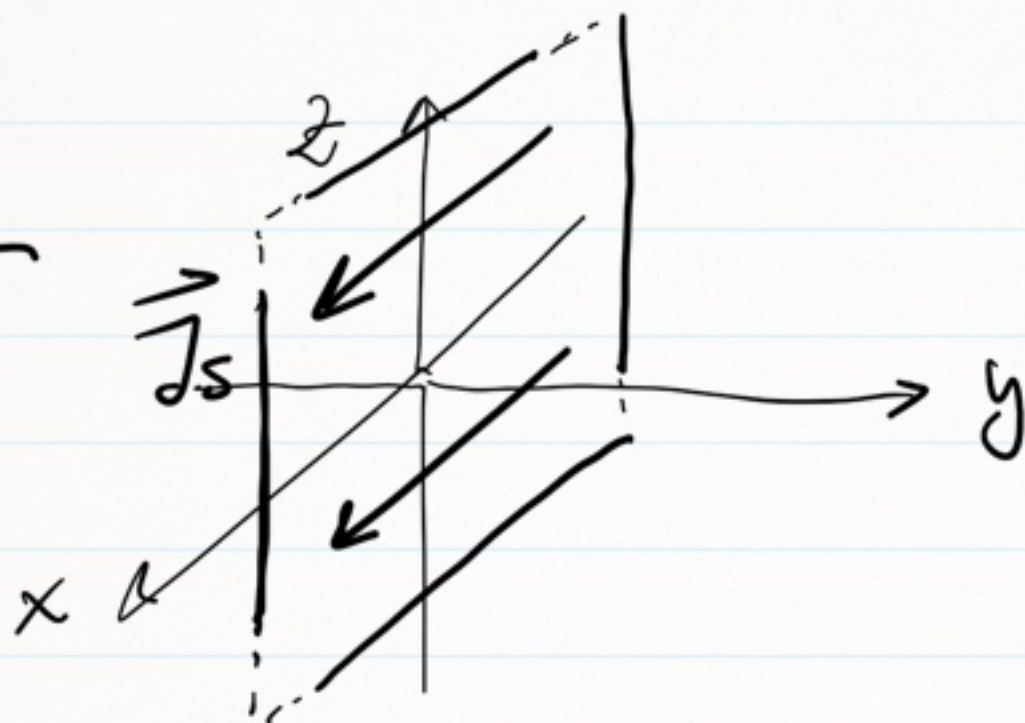
$$B = \frac{\mu_0 N I}{2\pi r}$$



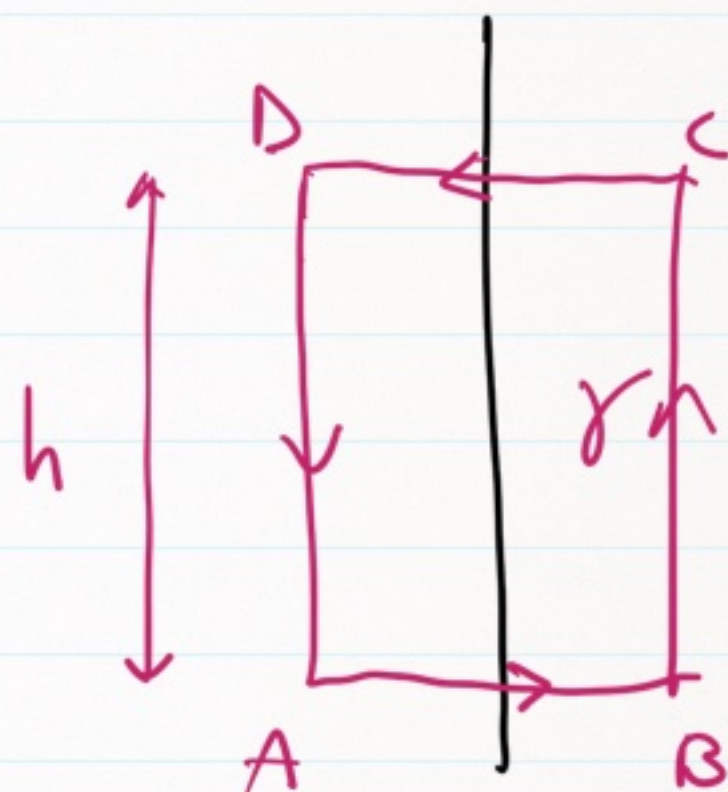
$$R_m = \frac{R_i + R_e}{2} = \frac{15 + 20}{2} = 17.5 \text{ cm}$$

$$B \Big|_{r=R_m} = \frac{2 \cdot 4\pi \cdot 10^{-7} \cdot 500 \cdot 0.8}{2\pi \cdot 0.175} = 6.4 \cdot 10^{-4} \text{ T}$$

Ex. 9



campo $\vec{B}$ risulta sempre
tangente all'interfaccia



circuito per
applicare Ampère
a $\vec{B}$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \cdot \underbrace{I_{\text{enc}}}_{J_S \cdot h}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \int_{BC+DA} \vec{B} \cdot d\vec{\ell} = B \cdot 2h$$

$$2Bh = \mu_0 J_S h \Rightarrow B = \frac{\mu_0 J_S}{2}$$

$$\begin{cases} \vec{B}_R = \frac{\mu_0 J_S}{2} \vec{u}_z \\ \vec{B}_L = -\frac{\mu_0 J_S}{2} \vec{u}_z \end{cases}$$

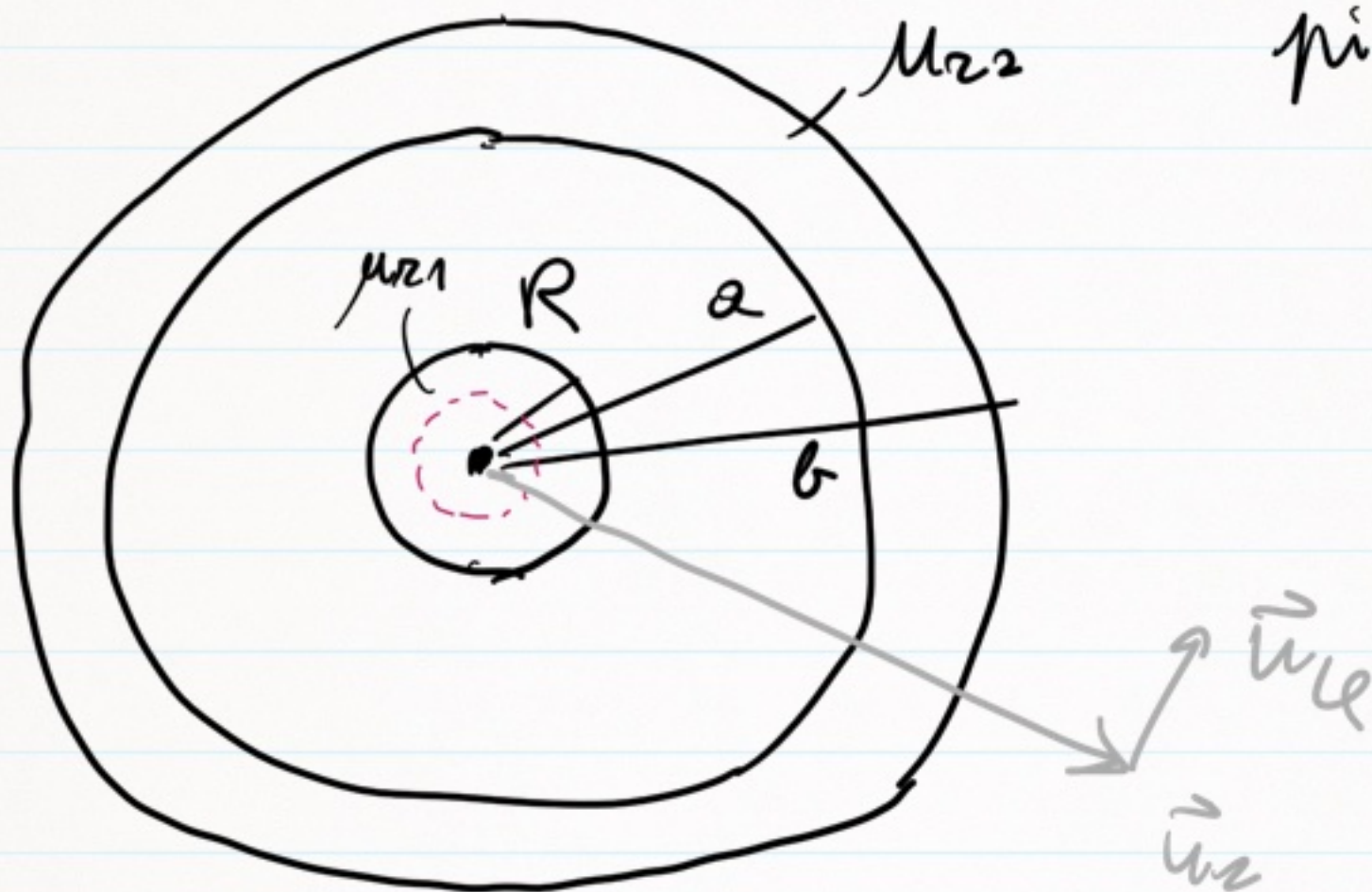
$$\Delta B = \underbrace{\mu_0 J_S}_{\uparrow}$$

$L \leftrightarrow R$

scheda 2

Ex. 1

I uscente del
piano



$$I = j \cdot \pi R^2, \quad I_2 = j \cdot \pi r^2$$

• $\vec{H} \quad (\vec{\nabla} \times \vec{H} = \vec{j})$

$\rightarrow 0 \leq r \leq R$

$\oint \vec{H} \cdot d\vec{\ell} = I_r$

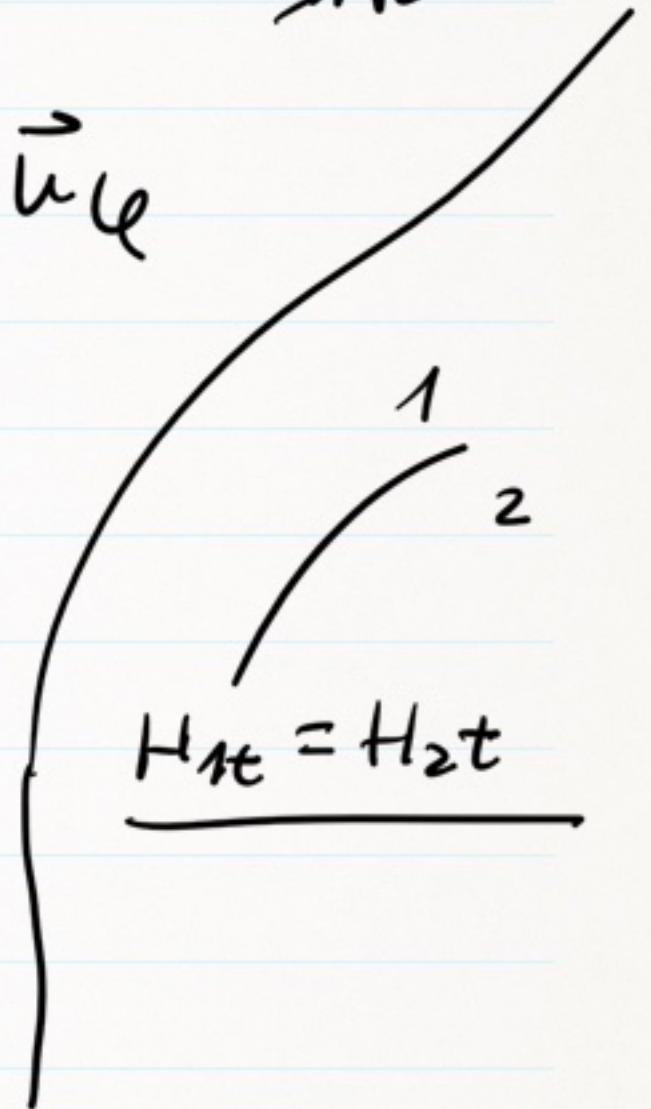
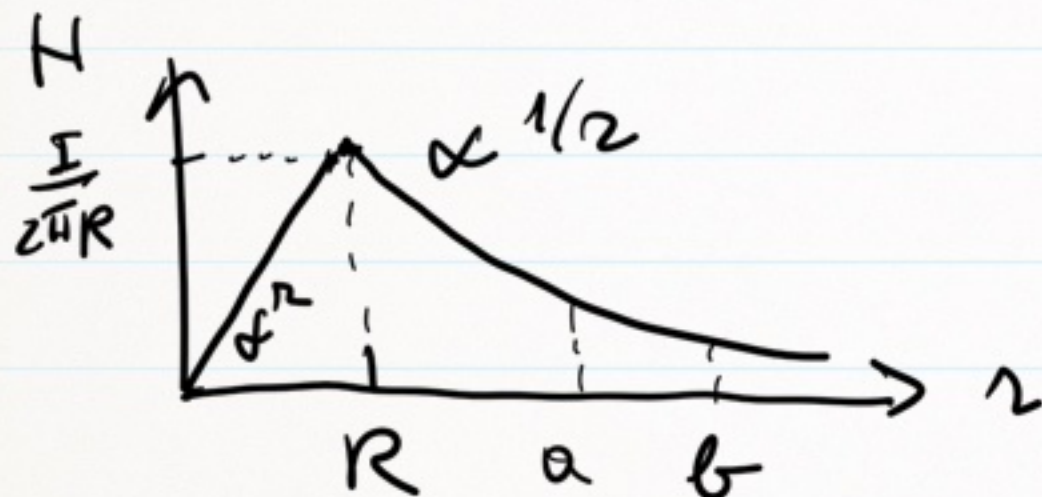
$H \cdot 2\pi r = I_r = I \cdot \frac{\pi r^2}{\pi R^2}$

$\vec{H} = \frac{I r}{2\pi R^2} \vec{u}_\phi$

$\rightarrow r > R$

$H \cdot 2\pi r = I$

$\vec{H} = \frac{I}{2\pi r} \vec{u}_\phi$



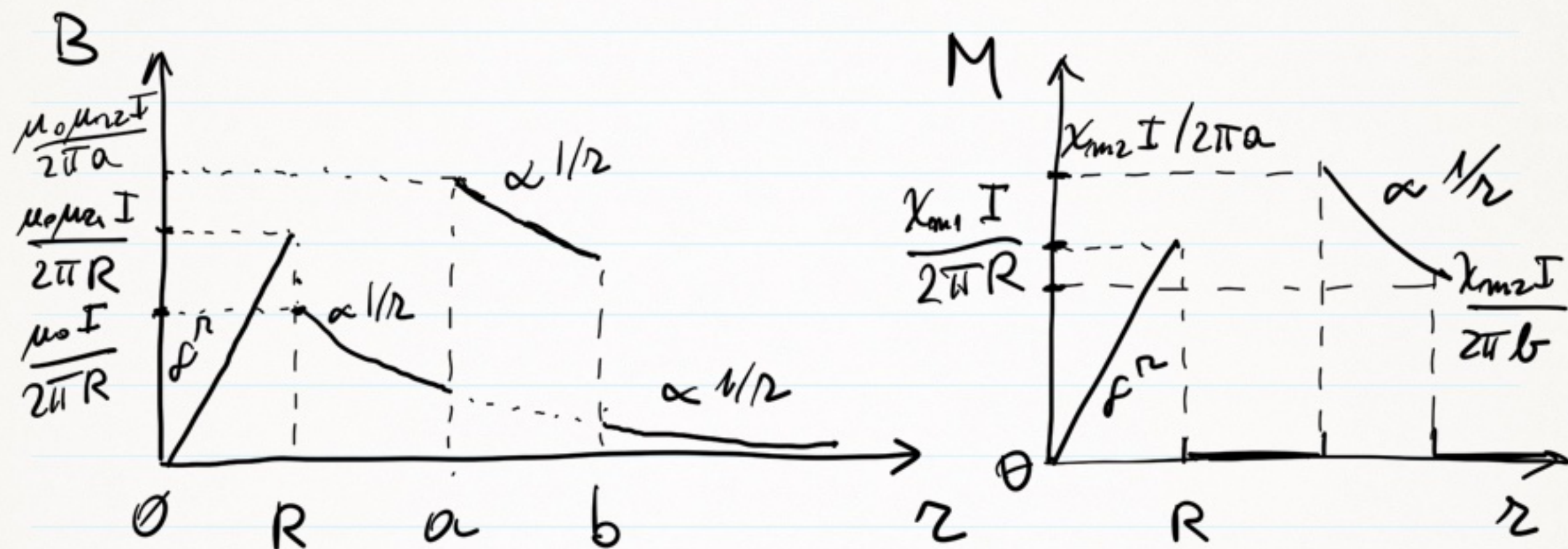
$$\begin{aligned}
 \cdot \vec{B} \quad \vec{B} &= \mu_0 (\vec{H} + \vec{M}) & \vec{M} &= \chi_m \cdot \vec{H} \\
 &= \mu_0 \underbrace{(1 + \chi_m)}_{\mu_r} \vec{H} = \mu_0 \mu_r \vec{H}
 \end{aligned}$$

$$0 \leq r \leq R^-: \quad \vec{B} = \mu_0 \mu_{r1} \vec{H} = \frac{\mu_0 \mu_{r1} I \cdot \underbrace{2\pi r}_{2\pi R^2}}{2\pi R^2} \vec{u}_\varphi$$

$$R^+ \leq r \leq a^-: \quad \vec{B} = \mu_0 \vec{H} = \frac{\mu_0 I}{\underbrace{2\pi r}_{2\pi R^2}} \vec{u}_\varphi$$

$$a^+ \leq r \leq b^-: \quad \vec{B} = \mu_0 \mu_{r2} \vec{H} = \frac{\mu_0 \mu_{r2} I}{\underbrace{2\pi r}_{2\pi R^2}} \vec{u}_\varphi$$

$$r > b^+: \quad \vec{B} = \mu_0 \vec{H} = \frac{\mu_0 I}{2\pi r} \vec{u}_\varphi$$



• $\vec{M}$

$$0 \leq r \leq R : \vec{M} = \chi_{m1} \cdot \vec{H} = \frac{\chi_{m1} I R}{2\pi R^2} \vec{u}_\varphi$$

$$a \leq r \leq b : \vec{M} = \chi_{m2} \cdot \vec{H} = \frac{\chi_{m2} I}{2\pi r} \vec{u}_\varphi$$

(b)

1) filo

$$\vec{J} = \frac{I}{\pi R^2} \vec{u}_z$$

$$\underline{\vec{J}_m} = \vec{V} \times \vec{M} = \chi_{m1} \vec{V} \times \vec{H} = \chi_{m1} \underline{\vec{J}}$$

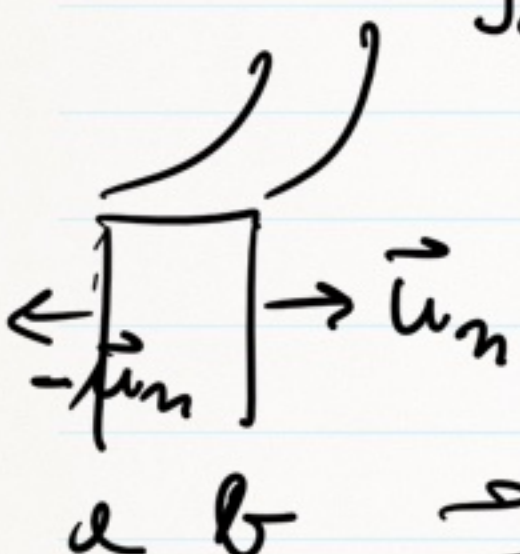
$$\underline{i_m} = J_m \cdot \pi R^2 = \chi_{m1} \cdot I$$

$$\begin{aligned} \vec{J}_{S,m} &= \vec{M} \times \vec{u}_n = \frac{\chi_{m1} I R}{2\pi R^2} \vec{u}_\varphi \times \vec{u}_n = \\ &\downarrow \\ @ R=R &= - \frac{\chi_{m1} I}{2\pi R} \vec{u}_z \end{aligned}$$

$$\lambda_{S,m} = \int_{S,m} \cdot 2\pi R = -\chi_{m1} \cdot I = -\lambda_m$$

(2) guaina

$$\vec{J}_m = \vec{0} \quad \text{perché } \vec{j} = \vec{0}$$



$$\vec{J}_{S,m} \Big|_{z=a} = - \vec{M}(z=a) \times \vec{u}_z$$

$$= + \frac{\chi_{m2} I}{2\pi a} \vec{u}_z$$

$$\vec{J}_{S,m} \Big|_{z=b} = \vec{M}(z=b) \times \vec{u}_z =$$

$$= - \frac{\chi_{m2} I}{2\pi b} \vec{u}_z$$

$$\underline{i_{s,m}|_a} = \int_{s,m} \cdot 2\pi a = \chi_{m2} \cdot I$$

$$i_{s,m}|_b = \int_{s,m} \cdot 2\pi b = -\chi_{m2} \cdot I = -\underline{i_{s,m}|_a}$$

ma se considero la regione : $a < r < b$

$$i_{tot} = I + \chi_{m2} I = (1 + \chi_{m2}) I = \mu_{r2} I$$



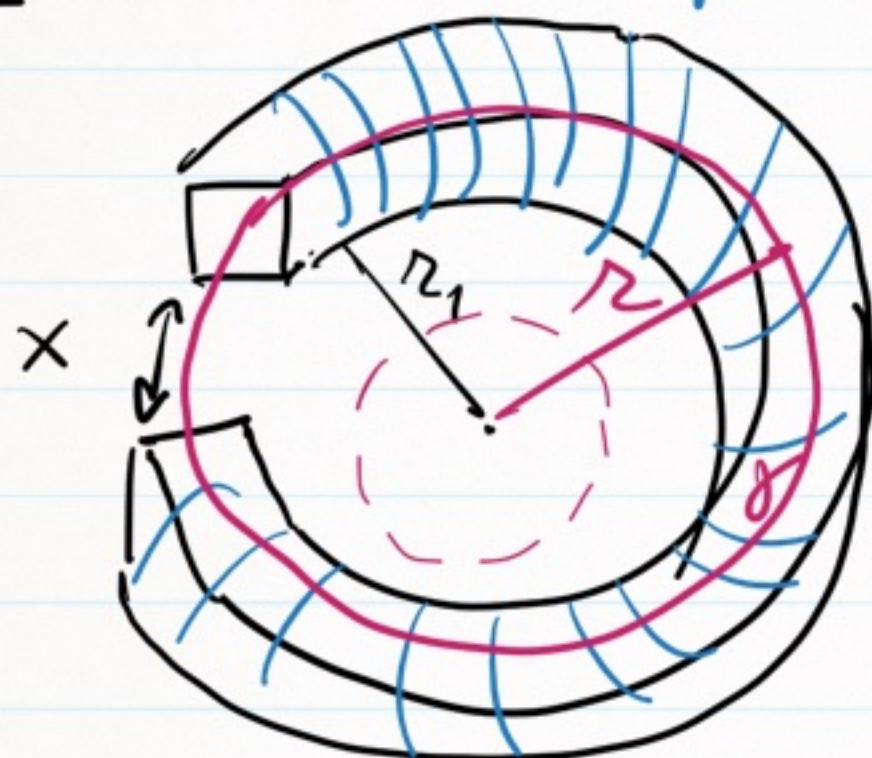
corrente totale equivalente sull'asse che

corrisponde al "rinforzo" di campo $\vec{B}$

visto nel grafico

Ex. 2

$N \neq \text{spine}$



$$B = B(r), H = H(r)$$

constant

$N \cdot I$

$(x \ll r_1)$

$$\oint \vec{H} \cdot d\vec{\ell} = \underbrace{I_{\text{enc}}}_{N \cdot I}$$

1: $\mu r \rightarrow 2\pi r - x$

2: $\text{aria (vasto)} \rightarrow x$

$$\oint \vec{H} \cdot d\vec{\ell} = \underline{H_1(r)} \cdot (2\pi r - x) + \underline{H_2(r)} \cdot x$$

$$B_1 = \mu_0 \mu_{r1} \textcircled{H_1}$$

$$B_2 = \mu_0 \textcircled{H_2}$$

$$H_1 = \frac{B}{\mu_0 \mu_{r1}}$$

$$H_2 = \frac{B}{\mu_0}$$

$$B \perp \text{taglios} \Rightarrow B_1 = B_2 = B$$

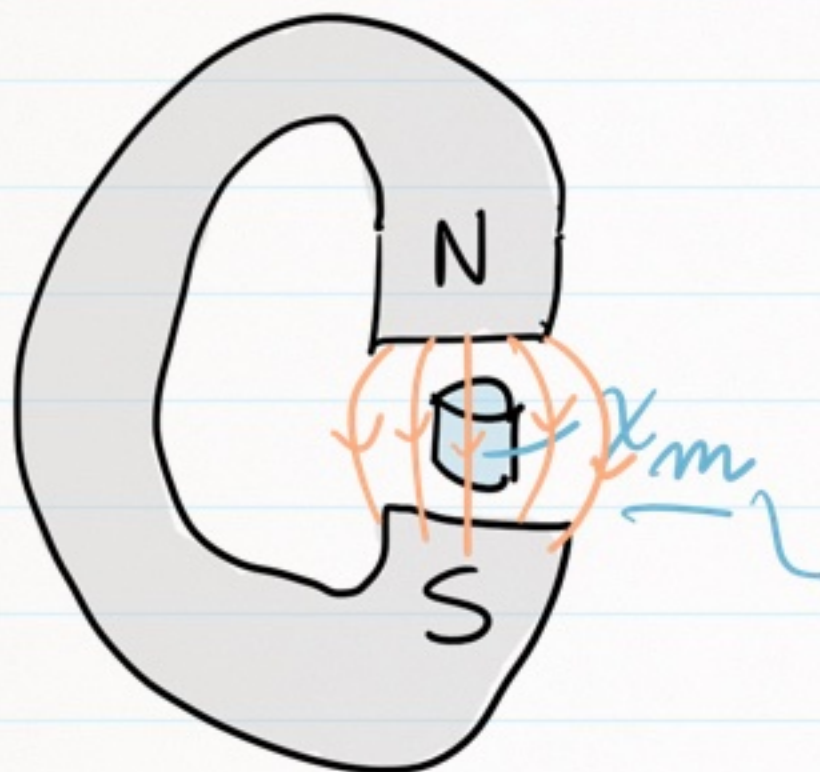
(conservazione
componente normale)

$$B = \frac{NI \mu_0 \mu_r}{\mu_r x + 2\pi r - x}$$

all'interno
del toroide

$$B = 0 \quad \text{altrove}$$

Ex. 3



- $F = F(r)$?

- $\vec{r}$ t.c. $F(\vec{r}) = F_{max}$?

$= -1.5 \cdot 10^{-5}$ DIAMAGNETICO

$$F = - \frac{\partial U_m}{\partial r}, \quad U_m = - \vec{m} \cdot \vec{B}, \quad \vec{m} = \vec{M} \cdot V$$

ricorda:

la magnetizzazione è una descrizione macroscopica che fornisce il momento magnetico per unità di volume

$$\vec{M} = \chi_m \cdot \vec{H} = \chi_m \cdot \vec{B} / \mu_0 \mu_r \approx \chi_m \cdot \vec{B} / \mu_0$$

$$\vec{m} = \frac{\chi_m V}{\mu_0} \vec{B}$$

$$\vec{F} = - \frac{\partial U_m}{\partial r} = - \frac{\partial}{\partial r} \left(- \frac{\chi_m V}{\mu_0} B^2 \right) =$$

$$= \frac{\chi_m V}{\mu_0} \cdot \frac{\partial}{\partial r} \left(B_0^2 \cdot e^{-2\alpha r^2} \right) =$$

$$= \frac{\chi_m V B_0^2}{\mu_0} \cdot (-4\alpha r) e^{-2\alpha r^2} \quad (\chi_m < 0)$$

$\Rightarrow \vec{F}$ respinge il cilindretto

$\wedge \vec{F} \rightarrow \vec{0}$ per $r \rightarrow \infty$

$$\frac{dF}{dr} = 0 : \left\{ - \frac{4\alpha \chi_m V B_0^2}{\mu_0} [r \cdot e^{-2\alpha r^2}] \right\}' = 0$$

$$e^{-2\alpha r^2} + r(-4\alpha r) e^{-2\alpha r^2} = 0$$

$$1 - 4\alpha r^2 = 0 \quad \Rightarrow \quad \bar{r} = \sqrt{\frac{1}{4\alpha}} = 3.53 \text{ cm}$$

$$F(\bar{r}) = 2.56 \cdot 10^{-6} \text{ N}$$